

Ch 06: Structures

The C Programming Language (2^{ed})

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Structures

Structure

A structure is a collection of one or more variables, possibly of different types, grouped together under a single name for convenient processing.

- Structures help to organize complicated data
 - Called “records”, “object”, “database entry or row”, etc.
- A `struct` defines a new type
 - Sometimes called a user defined data type
 - Essentially adding a new type to the language you can declare variables of and use
 - Access members **structure member operator** “.”

To declare a structure (don't forget training ;)):

```
struct point {  
    int x;  
    int y;  
};  
struct point pt;  
struct point maxpt = {320, 200 };  
pt.x = 320;  
pt.y = 200;
```



Pointers to Structures

- Can declare and use pointer to a struct, just like pointer to any defined type
- Special operator `->` will access and dereference a pointer to a structure.

See `kr06/ex01.c`

```
struct stuff {
    char *ptr;      // 8 bytes
    int num1;       // 4 bytes
    short num2;     // 2 bytes
    char byte;      // 1 bytes
    char bytes[8];  // 8 bytes
};

struct stuff s;
struct stuff *sp;

// a bunch of member access operations
s.ptr = "hello";
s.num1 = 0x11223344;
s.num2 = 0x4142; // 2 bytes 0x41 = ascii A, 0x42 = ascii B
s.byte = 'c';
s.bytes[0] = 'h';

sp = &s;
printf("num1: %d", sp->num1)
```



Summary

- A structure is a collection of one or more variables, of different types
 - Grouped together under a single name (record, row of a database, etc.)
 - User member operator . to access members
 - A structure will be a (byte aligned) sized allocation in memory
 - Can create arrays of structures, can pass structures to and return from functions
- Can declare and use pointer to a struct
 - Special operator -> performs both member access and dereference in single operation
 - Often we pass in a pointer to a structure buffer, to be filled in by system call



References

- Kernighan, B. W., & Ritchie, D. M. (1988). *The C programming language* (2nd). Pearson.
- TutorialsPoint. (2026). *C tutorial*. Retrieved August 17, 2026, from <https://www.tutorialspoint.com/cprogramming/index.htm>

